

Tribhuvan University  
Institute of Science and Technology  
B.Sc. CSIT Model Question 2075

1. \_\_\_\_\_ was present at the party.  
a) All but she                      b) All but herself  
c) All but her                      d) All but myself
2. Quinine is a drug \_\_\_\_\_ malaria.  
a) for                      b) to  
c) against                      d) on
3. During rainy season, the Koshi was  
a) overflow                      b) overflown  
c) overfly                      d) overflying
4. He is as stubborn as a \_\_\_\_\_.  
a) Child                      b) Bull  
c) Donkey                      d) Mule
5. Once you signed \_\_\_\_\_ the paper, you cannot go back.  
a) by                      b) in  
c) with                      d) out
6. The principal told the class \_\_\_\_\_ without permission.  
a) Not to enter                      b) Not to have entered  
c) Not entering                      d) Do not enter
7. I must go \_\_\_\_\_ before the shops are closed.  
a) Shopping                      b) To shop  
c) To shopping                      d) Shop
8. A horse neighs so as goats \_\_\_\_\_.  
a) Howl                      b) Roar  
c) Bleat                      d) Grunt
9. The teacher has authority \_\_\_\_\_ his students  
a) To                      b) Over  
c) With                      d) For
10. This is the place \_\_\_\_\_ we lived in  
a) Where                      b) Which  
c) When                      d) How
11. They have gone shopping, \_\_\_\_\_?  
a) Haven't they                      b) didn't they  
c) Had they                      d) Have they
12. Only after meeting him, \_\_\_\_\_ him well  
a) Did I know                      b) Had I know  
c) Had I known                      d) I know
13. Which of the following is not a synonym of splendid  
a) Superb                      b) Sordid  
c) Excellent                      d) Marvelous
14. Feminine of deer is  
a) Do                      b) Doa  
c) Doe                      d) Dode
15. \_\_\_\_\_ I open the present for you?  
a) Would                      b) Must  
c) Shall                      d) Do
16. He is \_\_\_\_\_ man of high esteem  
a) an                      b) a  
c) some                      d) none
17. He has a remarkable \_\_\_\_\_ for the arts.  
a) Talence                      b) Talency  
c) Talent                      d) Talents
18. Snow is falling \_\_\_\_\_ north of the mountains  
a) Into                      b) Over  
c) Off                      d) Down
19. I don't think he \_\_\_\_\_ come to the meeting tomorrow.  
a) Would                      b) Shall  
c) Used to                      d) Might
20. He is \_\_\_\_\_ one eyed man.  
a) the                      b) a  
c) an                      d) none
21. The study of the mountains  
a) geology                      b) geography  
c) chromatics                      d) orology
22. The opposite of the word pernicious is  
a) beneficial                      b) advantageous  
c) satisfactory                      d) fair
23. Grandma is \_\_\_\_\_ the visit of her grand children.  
a) Looking forward                      b) Looking into  
c) Looking for                      d) Looking forward to
24. Shyam has done this work and \_\_\_\_\_.  
a) So did I                      b) So am I  
c) So have I                      d) So has I
25. He \_\_\_\_\_ in plane crash  
a) Died                      b) Has died  
c) Have died                      d) In dying
26. The dimension ( $M^1L^2T^{-2}$ ) refers to a physical quantity that has unit:  
a) Joule                      b) Pascal  
c) Newton                      d) Watt
27. Which of the following is not a vector quantity  
a) Angular momentum                      b) Magnetic intensity  
c) Torque                      d) Energy
28. Whatever may be the direction of the two forces 6N and 2N acting on a body of mass 2Kg, the

minimum acceleration of the body cannot be less than

- a)  $4\text{m/s}^2$
- b)  $2\text{m/s}^2$
- c)  $2.5\text{m/s}^2$
- d)  $3\text{m/s}^2$

29. A body is moved through certain distance from rest under constant force. Then K.E. gained by the body is

- a) Directly proportional to its mass
- b) Inversely proportional to its mass
- c) Directly proportional to square root of mass
- d) Independent of its mass

30. A particle moves in a circle of radius 25cm at 2 rev/sec, the acceleration of the particle in  $\text{m/s}^2$  is

- a)  $\pi^2$
- b)  $4\pi^2$
- c)  $2\pi^2$
- d)  $8\pi^2$

31. The absolute zero temperature is

- a)  $-273^\circ\text{C}$
- b)  $-273\text{K}$
- c)  $-273.14^\circ\text{C}$
- d)  $-273.16^\circ\text{C}$

32. Melting point of ice

- a) Decrease with the decrease of pressure
- b) Increase with the increase of pressure
- c) Is independent of pressure
- d) Decrease with the increase of pressure

33. All gas at same temperature have same

- a) K.E
- b) Density
- c) RMS speed
- d) None of the above

34. The rate of loss of heat from a body depends on the

- a) Temperature of the body
- b) Excess temperature of body over the surrounding
- c) Thermal capacity of the body
- d) The temperature of the surrounding

35. The energy of molecular motion is expressed as:

- a) Friction
- b) Internal energy
- c) Temperature
- d) Potential energy

36. The velocity of sound in air is independent of change in

- a) Pressure
- b) Density
- c) Temperature
- d) Humidity

37. Sound waves having the following frequencies are audible to human beings

- a) 5c/s
- b) 27000c/s
- c) 5000c/s
- d) 50,000c/s

38. When a mirror is rotated through an angle  $30^\circ$  keeping incident ray constant then reflected ray is rotated through an angle

- a)  $25^\circ$
- b)  $60^\circ$

c)  $45^\circ$

d)  $90^\circ$

39. When light passes through glass slab

- a) Wavelength decreases
- b) Wavelength increases
- c) Velocity increases
- d) Frequency decreases

40. In an interference pattern minima are obtained when phase difference between interfering waves is

- a)  $\pi/2$
- b)  $2\pi$
- c)  $n\pi/2$
- d)  $(2n-1)\pi$

41. X-rays are produced by energy changes in

- a) Electrons close to the nucleus
- b) Electrons far from the nucleus
- c) Electrons and protons
- d) The nucleus

42. When a woolen sweater worn over a nylon shirt is removed. Sparking is observed due to

- a) Static electricity
- b) Current electricity
- c) None
- d) Both

43. The potential difference between the two charged parallel plates separated by 1mm is 100V, then the electric field produced is

- a)  $10^{-5}\text{V/m}$
- b)  $10^5\text{V/m}$
- c)  $10^3\text{V/m}$
- d)  $10^{-3}\text{V/m}$

44. The dielectric constant  $\epsilon_r$  is given by the relation

- a)  $\epsilon_r = \epsilon/\epsilon_0$
- b)  $\epsilon_r = \sqrt{(\epsilon/\epsilon_0)}$
- c)  $\epsilon_r = \epsilon/\epsilon_0$
- d) None

45. In a parallel plate capacitor force on each plate is

- a)  $q^2/\epsilon_0$
- b)  $q^2/A\epsilon_0$
- c)  $q^2/2A\epsilon_0$
- d) Zero

46. If the power of a heater is 1W and 1A of current is passed through it. Then find the resistance

- a)  $4.2\Omega$
- b)  $4200\Omega$
- c)  $1\Omega$
- d)  $0.1\Omega$

47. Wheatstone measures

- a) Potential
- b) emf
- c) Resistance
- d) Current

48. S.I. unit equivalent to the magnetic field Tesla(T) may be

- a)  $\text{Vsm}^2$
- b)  $\text{Vsm}^{-2}$
- c)  $\text{Vs}^{-1}\text{m}^2$
- d)  $\text{Vs}^{-1}\text{m}^{-2}$

49. Which of the following is used in the core of an electromagnet

- a) Soft magnet
- b) Soft steel
- c) Soft aluminum
- d) Soft zinc



50. A magnet is moved towards a coil through a certain distance slowly in one case and quickly in second case. The emf induced is  
 a) More in first case      b) More in second case  
 c) Equal in both cases      d) Zero in both case
51. Deliquescent salts can be used as  
 a) Oxidizing agent      b) Cleansing agent  
 c) Drying agent      d) Antiseptic
52. In the ideal gas equation  $PV=nRT$ , the value of gas constant depend only on  
 a) The pressure of the gas  
 b) The units of measurement  
 c) The nature of the gas  
 d) The volume of the gas
53. Equal weights of methane and hydrogen are mixed in an empty vessel at  $25^{\circ}\text{C}$ . The fraction of total pressure exerted by hydrogen is  
 a)  $1/2$       b)  $1/3$   
 c)  $1/9$       d)  $8/9$
54. Equivalent mass of a bivalent metal is 32.75 molecular mass of its chloride is  
 a) 68.25      b) 103.75  
 c) 101.00      d) 136.50
55. Two electrons in an orbital are distinguished by the quantum number  
 a)  $n$       b)  $l$   
 c)  $m$       d)  $s$
56. The bond in HCl molecule in the vapor state is an example of  
 a) Non polar bond      b) Polar covalent bond  
 c) Ionic bond      d) Pure covalent bond
57. Electron affinity is maximum in the case of  
 a) Cl      b) F  
 c) B      d) N
58. Which of the following species can act both as an acid and as a base  
 a)  $\text{H}_3\text{O}^+$       b)  $\text{HSO}_4^-$   
 c)  $\text{HCO}_3^-$       d)  $\text{Cl}^-$
59. In which of the following reaction is  $\text{H}_2$  acting as oxidizing agent?  
 a)  $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$       b)  $\text{Ca} + \text{H}_2 \rightarrow \text{CaH}_2$   
 c)  $\text{H}_2 + \text{Cl}_2 \rightarrow \text{HCl}$       d)  $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$
60. When 100 ml of 1M solution of  $\text{H}_2\text{SO}_4$  and 20 ml of 5M NaOH are mixed, the resulting solution would be  
 a) Acidic      b) Weakly alkaline  
 c) Strongly alkaline      d) Neutral
61. Which pure substance will not conduct electricity?  
 a) Liquefied HCl      b) Molten NaCl  
 c) Molten KOH      d) Liquid Na
62. The oxidation of  $\text{SO}_2$  by  $\text{O}_2$  to form  $\text{SO}_3$  is an exothermic reaction. Production of  $\text{SO}_3$  will be maximum  
 a) If temperature is raised  
 b) If temperature is decreased  
 c) If concentration of  $\text{SO}_2$  is decreased  
 d) If concentration of  $\text{O}_2$  is decreased
63. The process which is catalyzed by one of the product is known as  
 a) Acid catalysis      b) Positive catalysis  
 c) Anti-catalysis      d) Auto-catalysis
64. In electrolysis of  $\text{CuSO}_4$  solution between Pt-electrodes  
 a)  $\text{O}_2$  is liberated at anode  
 b)  $\text{H}_2$  is liberated at cathode  
 c)  $\text{SO}_2$  is liberated at anode  
 d)  $\text{Cu}^{++}$  is discharged at anode
65. Precipitation takes place when the ionic product is  
 a) Zero      b) Unity  
 c) Less than solubility product  
 d) More than solubility product
66. Living in the atmosphere of Carbon monoxide is dangerous because it  
 a) Combines with hemoglobin and makes it incapable to absorb  $\text{O}_2$   
 b) Dries up blood  
 c) Combines with oxygen present inside to form  $\text{CO}_2$   
 d) Reduces organic matter of tissues
67. Yellow color of  $\text{HNO}_3$  is due to the presence of  
 a)  $\text{N}_2\text{O}$       b) NO  
 c)  $\text{NO}_2$       d)  $\text{N}_2\text{O}_5$
68. When  $\text{SO}_2$  is passed through acidified solution of  $\text{H}_2\text{S}$   
 a)  $\text{H}_2\text{SO}_3$  is formed      b)  $\text{H}_2\text{SO}_2$  is formed  
 c)  $\text{H}_2\text{SO}_4$  is formed      d) S is precipitated
69. The compound having highest boiling point  
 a) HF      b) HCl  
 c) HBr      d) HI
70. Which of the following processes involves the smelting process?  
 a)  $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O} \rightarrow \text{Al}_2\text{O}_3 + 2\text{H}_2\text{O}$

- b)  $\text{ZnCO}_3 \rightarrow \text{Zn} + \text{CO}_2$   
 c)  $2\text{PbS} + \text{O}_2 \rightarrow 2\text{PbO} + 2\text{SO}_2$   
 d)  $\text{Fe}_2\text{O}_3 + 3\text{C} \rightarrow 2\text{Fe} + 3\text{CO}$
71. In Down's process for manufacture of metallic sodium a small amount of  $\text{CaCl}_2$  is added to  
 a) Increase the electrical conductivity  
 b) Lower melting point of  $\text{NaCl}$   
 c) Increase the temperature of electrolysis  
 d) Stabilize the metallic sodium
72. Which of the following is malachite ore  
 a)  $\text{Cu}_2\text{S}$  b)  $\text{Cu}_2\text{O}$   
 c)  $\text{CuCO}_3$  d)  $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$
73. Cosmetic powders and zinc ointments contain  
 a)  $\text{ZnO}$  b)  $\text{ZnCO}_3$   
 c)  $\text{Zn}(\text{NO}_3)_2$  d)  $\text{ZnSO}_4$
74. Rusting of iron in moist air involves  
 a) Loss of electrons by Fe  
 b) Gain of electrons by Fe  
 c) Dehydration of Fe d) Hydration of Fe
75. Aromaticity of benzene is due to  
 a) Ring b) Hyper conjugation  
 c) Three double bonds  
 d) Delocalization of electrons
76. If  $|x-2|=5$ , then x is in  
 (a) {3,5} (b) {2,∞}  
 (c) {2,5} (d) {-3,7}
77.  $\mathbf{A-B} =$   
 (a)  $\mathbf{B-A}$  (b)  $\overline{\mathbf{A}} - \overline{\mathbf{B}}$   
 (c)  $\overline{\mathbf{B}} - \overline{\mathbf{A}}$  (d)  $\overline{\mathbf{B}} - \mathbf{A}$
78. If  $f(x)=2x-3$ , then  $f^{-1}(x) =$   
 (a)  $(x+3)/2$  (b)  $1/(2x-3)$   
 (c)  $1/(3-2x)$  (d)  $1/(2x+3)$
79. The period of the function  $f(x) = \sin(45+5)$  is  
 (a)  $2\pi$  (b)  $\pi/2$   
 (c)  $\pi$  (d)  $-\pi$
80. The fourth, seventh and tenth terms of a G.P. are l, m, n respectively, then  
 (a)  $ln = m^2$  (b)  $l^2 = m^2 + n^2$   
 (c)  $l^2 = mn$  (d)  $n^2 = lm$
81. The nth term of the series:  $1.3+3.5+5.7+\dots$  is  
 (a)  $(n+1)(n+2)$  (b)  $(n+1)(n+3)$   
 (c)  $4n^2-1$  (d)  $n(n-1)$
82. The value of  $\sin(\cos^{-1}x)$  is  
 (a) x (b)  $1-x^2$   
 (c)  $1+x^2$  (d)  $\sqrt{1-x^2}$
83. If  $\operatorname{cosec}^2 x - 2 = 0$ , then x is  
 (a)  $\pm \pi/6$  (b)  $\pm \pi/4$   
 (c)  $n\pi \pm \pi/4$  (d)  $n\pi + \pi/2$
84. If  $\begin{vmatrix} k & 1 & 0 \\ 2 & 0 & k \\ 0 & 2 & -1 \end{vmatrix} = 0$ , then k =  
 (a) 1 (b) -1  
 (c) 0 (d)  $\pm 1$
85. The value of x for which  $\mathbf{A} = \begin{bmatrix} 6 & x-2 \\ 3 & x \end{bmatrix}$  has no inverse is  
 (a) 0 (b) 1  
 (c) -2 (d) 2
86. If the system of equation has zero solutions only then the solution is  
 (a) non trivial (b) trivial  
 (c) unique (d) particular
87.  $(1+i)^6 + (1-i)^6 =$   
 (a) 0 (b) i  
 (c) 1 (d) -1
88.  $\left(\frac{2i}{1+i}\right)^2 =$   
 (a) -i (b)  $1+i$   
 (c) 2i (d) -2i
89. If one root of  $5x^2 + 2x - k = 0$  is the reciprocal of the other than k =  
 (a) -5 (b) 5  
 (c) 2/5 (d) 5/2
90. The equation  $x^2 - 2xy + y^2 + 3x - 2 = 0$  represents:  
 (a) hyperbola (b) parabola  
 (c) ellipse (d) circle
91. The area of the triangle with vertices (1,-1), (-1,1) and (-1,-1) is  
 (a)  $\frac{1}{2}$  (b) -2  
 (c) 2 (d) 0
92.  $\lim_{n \rightarrow \infty} \frac{1+2+\dots+n}{n^2} =$   
 (a)  $\frac{1}{2}$  (b) 0  
 (c)  $1/n$  (d) n
93.  $\lim_{x \rightarrow 0} \frac{5x^2 + 4x \tan x}{x^2} =$   
 (a) 9 (b) 0



- (c) 5 (d) 4
94. If  $y = \sqrt{x + \sqrt{x + \sqrt{x + \dots}}}$  then  $\frac{dy}{dx} =$   
 (a)  $2\sqrt{x}$  (b)  $\sqrt{x^3}$   
 (c)  $\frac{\sqrt{x}}{2y-1}$  (d)  $\frac{1}{2y-1}$
95. If  $y = e^{\sqrt{ax^2+b}}$  then  $dy/dx =$   
 (a)  $\frac{axy}{\sqrt{ax^2+b}}$  (b)  $\frac{ax}{\sqrt{ax^2+b}} e^{\sqrt{ax^2+b}}$   
 (c)  $\frac{x}{\sqrt{ax^2+b}}$  (d)  $e^{\sqrt{ax^2+b}}$
96.  $\int \frac{dx}{1-x^2} =$   
 (a)  $\cos^{-1}x + c$  (b)  $\frac{1}{2} \ln \left| \frac{1-x}{1+x} \right| + c$   
 (c)  $\sin^{-1}x + c$  (d)  $\frac{1}{2} \ln \left| \frac{1+x}{1-x} \right| + c$
97.  $\int_1^2 \frac{\sin(\ln t)}{t} dt =$   
 (a)  $1 - \cos(\ln 2)$  (b)  $\cos 2$   
 (c)  $\cos(\ln 2)$  (d)  $1 + \cos(\ln 2)$
98. The critical points for  $f(x) = x^3 - 3x$  are:  
 (a)  $\pm 3$  (b)  $\pm \sqrt{3}$   
 (c)  $\pm 1$  (d)  $0, \pm \sqrt{3}$
99. The area bounded by the x-axis and the curve  $y = x^3$  and ordinates of  $x = 2$  and  $x = 4$  is  
 (a) 60 sq. units (b) 256 sq. units  
 (c) 240 sq. units (d) 272 sq. units
100. If  $\log_a 81 = 4$  then the value of  $a =$   
 (a) 4 (b) 5  
 (c) 3 (d) -4

#### English Answer:

1.a 2.c 3.d 4.a 5.b 6.a 7.a 8.c 9.b 10.a 11.a  
 12.a 13.b 14.c 15.c 16.b 17.c 18.b 19.a 20.b 21.d  
 22.a 23.d 24.c 25.a

#### Physics Answer:

26.a 27.d 28.b 29.d 30.a 31.d 32.d 33.d 34.b  
 35.b 36.a 37.c 38.b 39.a 40.d 41.d 42.b 43.b 44.c  
 45.c 46.c 47.c 48.b 49.b 50.b

#### Chemistry Answer:

51.b 52.b 53.d 54.d 55.d 56.c 57.a 58.b 59.b 60.a  
 61.c 62.b 63.d 64.A 65.d 66.a 67.d 68.c 69.d  
 70.d 71.B 72.d 73.b 74.a 75.d

#### Mathematics Answer:

76.d 77.c 78.a 79.a 80.a 81.c 82.d 83.c  
 84.b 85.c 86.b 87.a 88.b 89.a 90.d 91.c 92.a  
 93.a 94.d 94.b 96.b 97.a 98.c 99.a 100.c

## FORMULA BOOK SECTION